Real-Time Occupancy Detection System for Energy Efficiency

1. Problem Statement

In modern buildings, educational institutions, industries, and offices, a significant amount of electrical energy is wasted because lighting systems remain switched ON even when the area is unoccupied. Traditional lighting systems rely on manual operation, which often results in unnecessary power consumption due to human negligence. With the increasing demand for energy conservation and sustainable resource utilization, automated occupancy-based control systems have become an important area of research and development.

Real-time occupancy detection systems use sensors and microcontrollers to determine whether a space is occupied and automatically control electrical appliances accordingly. Such systems help reduce energy wastage, lower electricity costs, and improve overall operational efficiency. These solutions are widely used in smart homes, offices, classrooms, warehouses, and industrial facilities.

This project presents a Real-Time Occupancy Detection System for Energy Efficiency using an Arduino Uno microcontroller and a Passive Infrared (PIR) sensor. The PIR sensor detects human motion within a specified range and sends a signal to the Arduino Uno. Based on this input, the Arduino automatically turns the lighting system ON when occupancy is detected and switches it OFF after a predefined period of inactivity.

The proposed solution provides a simple, low-cost, and effective method for reducing energy wastage in indoor environments. The project demonstrates the practical application of embedded systems, sensor technology, and automation principles in developing energy-efficient solutions for homes, classrooms, offices, and laboratories.

Problem Definition / Challenge

Energy conservation has become a critical concern due to the increasing demand for electricity and the growing emphasis on sustainable development. In many homes, offices, classrooms, laboratories, and industrial facilities, lights and other electrical appliances are often left switched ON even when the area is unoccupied. This unnecessary operation results in significant energy wastage, increased electricity costs, and reduced operational efficiency.

Traditional lighting systems depend on manual control, making them prone to human error and negligence. Occupants may forget to switch OFF lights when leaving a room, leading to avoidable power consumption. Existing advanced solutions such as camera-based monitoring systems and IoT-enabled smart building technologies can address this issue but are often expensive, complex to deploy, and may raise privacy concerns.

The challenge is to develop a simple, cost-effective, and reliable system capable of detecting human occupancy in real time and automatically controlling lighting based on occupancy status. The system should minimize energy wastage, require minimal human intervention, and be easy to implement using readily available hardware components.

To address this challenge, this project proposes a Real-Time Occupancy Detection System using a PIR sensor and Arduino Uno. The system continuously monitors human presence and automatically controls lighting, thereby improving energy efficiency and promoting smart automation practices.

Literature / Market Review of Existing Solutions and Their Limitations

Several occupancy detection and lighting control systems have been developed to reduce energy wastage in residential, commercial, and industrial environments. Some commonly used approaches are discussed below.

1. Manual Switch-Based Lighting Systems

Manual lighting systems require users to switch lights ON and OFF as needed. While they are simple and inexpensive, they rely entirely on human behavior and often result in lights being left ON unnecessarily, causing energy wastage.

2. Timer-Based Lighting Control Systems

Timer-based systems operate lights according to predefined schedules. Although they provide basic automation, they cannot detect actual room occupancy and may keep lights ON even when the area is vacant.

3. Ultrasonic Sensor-Based Occupancy Detection

Ultrasonic sensors detect movement using sound waves and can operate in low-light environments. However, they are more susceptible to environmental interference and generally consume more power than PIR-based systems.

4. Camera-Based Occupancy Detection Systems

Camera-based systems use image processing and artificial intelligence to detect human presence. These systems offer high accuracy but involve higher costs, greater computational requirements, and privacy concerns.

5. IoT-Based Smart Building Systems

IoT-enabled systems provide remote monitoring and advanced automation capabilities through network connectivity. However, they increase system complexity, implementation cost, and dependence on internet infrastructure.

System	Cost	Privacy	Accuracy	Complexity
Manual	Low	High	Low	Low
Timer	Low	High	Medium	Low
PIR	Low	High	Medium	Low
Camera	High	Low	High	High
IoT	Medium	Medium	High	High

Identified Research Gap

Existing occupancy detection solutions either suffer from high cost, implementation complexity, or privacy concerns, while simpler systems fail to respond to actual occupancy conditions. Therefore, there is a need for a low-cost, easy-to-implement, and reliable occupancy detection system that can automatically control lighting based on real-time human presence without requiring expensive hardware or internet connectivity.

Proposed Solution

To address the challenges associated with energy wastage in indoor environments, this project proposes a Real-Time Occupancy Detection System using an Arduino Uno microcontroller and a Passive Infrared (PIR) sensor.

The PIR sensor continuously monitors the surroundings for human motion and sends occupancy information to the Arduino Uno. When motion is detected, the Arduino automatically activates the connected lighting system represented by an LED. If no motion is detected for a predefined period, the lighting system is switched OFF to conserve energy.

The proposed solution provides a simple and cost-effective method of occupancy-based lighting control. Compared to manual lighting systems, it reduces human intervention and minimizes unnecessary electricity consumption. Unlike camera-based occupancy detection systems, the proposed system avoids privacy concerns and requires significantly lower hardware and computational resources.

The solution is suitable for classrooms, offices, laboratories, meeting rooms, and residential environments where energy conservation is important.

2. Scope of the Solution

The proposed Real-Time Occupancy Detection System is designed to detect human occupancy in indoor environments and automatically control lighting to improve energy efficiency.

The primary scope of the solution is to identify human presence within a monitored area using a PIR sensor and activate the lighting system only when occupancy is detected. When no motion is detected for a predefined duration, the Arduino Uno automatically switches OFF the lighting load, thereby reducing unnecessary energy consumption.

The system is intended for deployment in classrooms, offices, laboratories, meeting rooms, warehouses, and residential environments where energy conservation and automated lighting control are beneficial.

The proposed solution provides the following capabilities:

• Real-time occupancy detection using a PIR sensor.

• Automatic control of lighting based on occupancy status.

• Reduced energy wastage through occupancy-based automation.

• Low-cost implementation using commercially available components.

• Simple architecture suitable for educational and prototype applications.

Limitations

• PIR sensors detect motion rather than continuous human presence.

• Occupants remaining completely stationary for extended periods may not be detected.

• Detection performance depends on sensor placement and environmental conditions.

• The prototype is designed for single-room monitoring and may require additional sensors for larger spaces.

Future Scope

The proposed system can be enhanced by integrating additional sensors, wireless communication modules, and smart control mechanisms. Future versions may support automated control of multiple electrical appliances, energy consumption monitoring, occupancy analytics, and integration with smart home systems. The system can also be expanded for large-scale deployment in educational institutions, offices, and commercial buildings to improve overall energy efficiency.

Objectives

The primary objective of the proposed Real-Time Occupancy Detection System is to improve energy efficiency by automatically controlling lighting based on real-time occupancy detection.

The specific objectives are:

• To detect human occupancy using a PIR motion sensor in real time.

• To automatically switch ON the lighting system when occupancy is detected.

• To automatically switch OFF the lighting system after a predefined period of inactivity.

• To reduce unnecessary electricity consumption and improve energy efficiency.

• To demonstrate a low-cost automation solution using Arduino Uno.

• To apply embedded system concepts in a practical energy-saving application.

Boundaries of the Solution

The proposed system is designed as a prototype for indoor occupancy monitoring and automated lighting control. The scope of implementation is limited to the following boundaries:

• The system detects occupancy based on motion using a PIR sensor.

• The prototype controls a single lighting load represented by an LED.

• The system is intended for indoor environments such as classrooms, offices, laboratories, meeting rooms, and homes.

• The prototype focuses only on occupancy-based lighting control.

• The system does not control HVAC systems, security systems, or other building services.

• The prototype is designed for proof-of-concept demonstration and educational purposes.

• The system does not perform facial recognition, identity tracking, or camera-based monitoring, thereby preserving user privacy.

Solution Aims

The proposed Real-Time Occupancy Detection System aims to reduce energy wastage by automatically controlling lighting based on real-time occupancy detection. The system utilizes a PIR sensor to detect human motion and an Arduino Uno microcontroller to process sensor inputs and manage lighting operation.

The solution aims to provide a low-cost, reliable, and energy-efficient alternative to traditional manual lighting systems. By reducing the need for manual switching, the system minimizes unnecessary electricity consumption and promotes efficient utilization of energy resources.

Furthermore, the project aims to demonstrate the practical application of embedded systems and sensor-based automation in developing energy-saving solutions for indoor environments such as classrooms, offices, laboratories, and residential buildings.

Constraints and Limitations of the Proposed Solution

While the proposed Real-Time Occupancy Detection System provides an effective and low-cost solution for occupancy-based lighting control, certain constraints and limitations must be considered.

1. **Motion-Based Detection**

The system uses a Passive Infrared (PIR) sensor, which detects changes in infrared radiation caused by movement.

Reasoning:

If an occupant remains stationary for an extended period, the PIR sensor may not detect their presence. As a result, the system may incorrectly assume that the room is unoccupied and switch OFF the lighting system.

2. **Limited Detection Range and Coverage**

The performance of the PIR sensor is limited by its detection range and field of view.

Reasoning:

Large rooms, partitions, obstacles, or improper sensor placement may create blind spots where human movement cannot be detected effectively.

3. **Prototype-Scale Implementation**

The current implementation is designed as a proof-of-concept prototype for a single room.

Reasoning:

Monitoring larger spaces would require additional sensors and control mechanisms, increasing system complexity and cost.

4. **Environmental Influences**

PIR sensors may be affected by environmental conditions such as heat sources and rapid temperature changes.

Reasoning:

Strong heat-emitting objects or improper installation locations may reduce detection accuracy and lead to false triggering or missed detections.

5. **Lighting-Focused Automation**

The prototype is primarily designed to control lighting based on occupancy status.

Reasoning:

Other building systems such as HVAC, security systems, and access control are not integrated into the current implementation and would require further expansion of the system architecture.

6. **Privacy-Preserving Design Trade-Off**

The system intentionally avoids camera-based monitoring to protect user privacy.

Reasoning:

While this eliminates privacy concerns and reduces implementation cost, it also limits advanced capabilities such as occupant identification, people counting, and detailed occupancy analytics available in vision-based systems.

Justification of the Proposed Solution

The proposed Real-Time Occupancy Detection System has been selected as an effective solution for addressing energy wastage caused by unnecessary lighting usage in indoor environments. The system combines occupancy detection and automated lighting control to provide a practical and energy-efficient approach for energy conservation.

One of the primary reasons for selecting this solution is its ability to reduce electricity consumption through occupancy-based automation. By ensuring that lighting systems operate only when a space is occupied, the system minimizes unnecessary energy usage and contributes to improved energy efficiency.

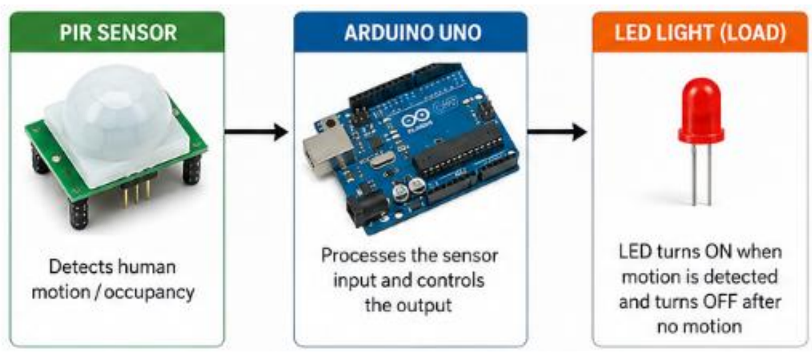
The Arduino Uno microcontroller was selected because it is inexpensive, widely available, easy to program, and highly suitable for educational and embedded system applications. Its compatibility with a wide range of sensors makes it an ideal platform for implementing occupancy-based automation systems.

A PIR sensor has been chosen for occupancy detection because it is inexpensive, reliable, energy-efficient, and easy to integrate with Arduino-based systems. Compared to camera-based occupancy detection systems, PIR sensors require significantly lower computational resources and do not raise privacy concerns related to image capture and storage.

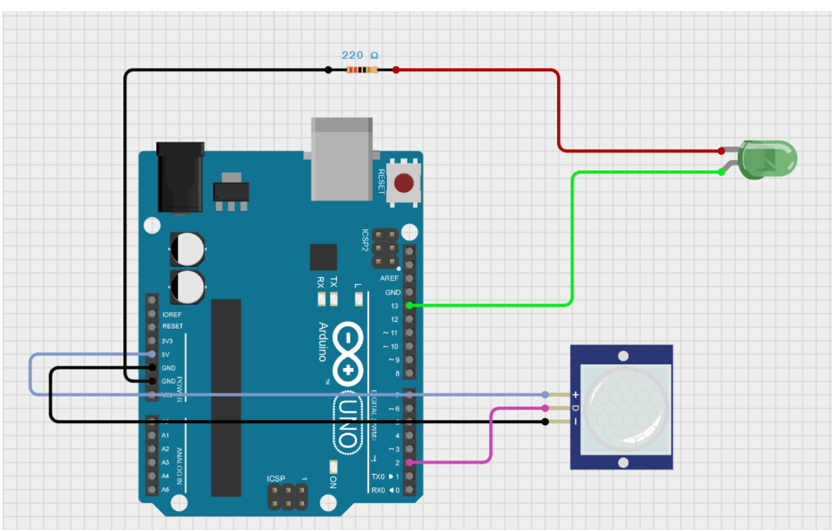
The proposed solution is simple, scalable, and easy to implement. Although the current implementation focuses on a single-room prototype, the concept can be extended for larger environments with additional sensors and control units.

In addition, the solution supports the objectives of sustainable energy management by reducing unnecessary power consumption. By combining low-cost hardware and automated control, the proposed system provides a practical foundation for future smart automation applications while remaining suitable for educational and prototype development purposes.

3. Block diagram



4. Circuit diagram



5. Required Components to Develop Solution

5.1. Hardware Components

S.No	Component	Quantity	Purpose
1	Arduino Uno	1	Main microcontroller used for processing sensor inputs and controlling the output
2	HC-SR501 PIR Motion Sensor	1	Detects human motion and occupancy
3	LED	1	Represents the lighting system
4	220Ω Resistor	1	Limits current through the LED
5	Breadboard	1	Circuit prototyping and testing
6	Jumper Wires	As required	Electrical connections between components
7	USB Cable	1	Programming and powering the Arduino Uno

5.2. Software Components

S.No	Software	Purpose
1	Arduino IDE	Writing, compiling, and uploading code to Arduino Uno
2	EasyEDA	Gerber files creation
3	Cirkit Designer IDE	Circuit simulation and testing
4	GitHub	Project documentation and version control

5.3. IDE Used

Arduino IDE (Version 2.x)

The Arduino Integrated Development Environment (IDE) is used to develop, compile, debug, and upload the source code to the Arduino Uno microcontroller. It provides an easy-to-use platform for writing and testing embedded system applications.

5.4. Libraries and Frameworks

Library / Framework	**Purpose**
Arduino Core Library	Provides basic functions for Arduino programming such as digital input/output operations, timing functions, and serial communication

5.5. Hardware Specifications

Arduino Uno

• Microcontroller: ATmega328P

• Operating Voltage: 5V

• Input Voltage (Recommended): 7V – 12V

• Digital I/O Pins: 14

• Analog Input Pins: 6

• Clock Speed: 16 MHz

• Flash Memory: 32 KB

• SRAM: 2 KB

• EEPROM: 1 KB

• USB Interface for Programming

HC-SR501 PIR Motion Sensor

• Operating Voltage: 5V

• Detection Range: Up to 7 meters

• Detection Angle: Approximately 120°

• Adjustable Sensitivity and Delay

• Low Power Consumption

LED

• Operating Voltage: 2V – 3V

• Used as an indicator for occupancy-based lighting control

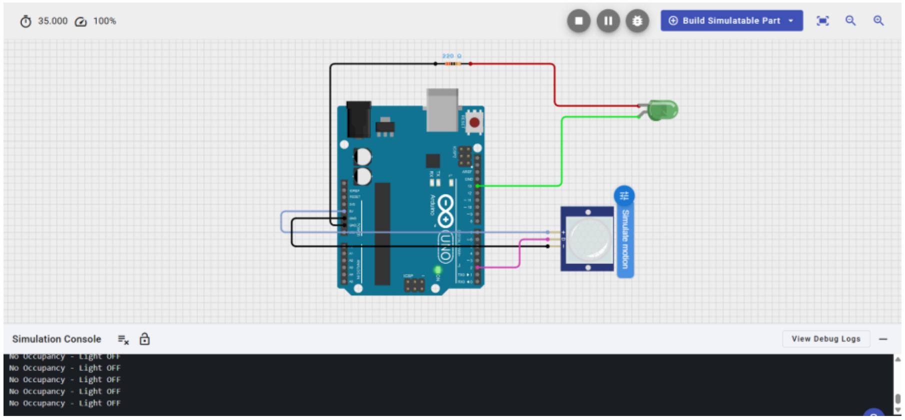
5.6. Specialized Equipment

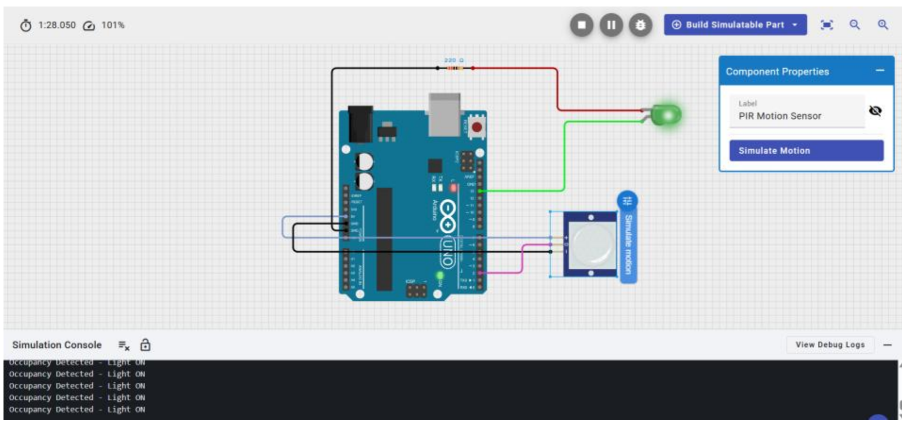
• Laptop/PC for programming and simulation

• Breadboard and jumper wires for circuit implementation

• USB cable for Arduino Uno programming and power supply

6. Simulated Circuit





Online simulation link: https://app.cirkitdesigner.com/project/fc8c1c20-c3ee-4f36-a7d3-30ea475516cd

7. Code for the Solution

Programming Language: C++ (Arduino)

```
/* Real-Time Occupancy Detection System for Energy Efficiency

  This project uses a PIR motion sensor to detect human occupancy

  and automatically control a lighting system. When motion is detected,

  the light turns ON. If no motion is detected for 10 seconds,

  the light turns OFF to save energy.

  Components:

  - Arduino Uno

  - PIR Motion Sensor (HC-SR501)
```

```
  - LED
  - 220Ω Resistor

  Connections:
  PIR Sensor:
    VCC -> 5V
    GND -> GND
    OUT -> Digital Pin 2
  LED:
    Anode (+) -> Digital Pin 13
    Cathode (-) -> 220Ω Resistor -> GND
*/

#define PIR_PIN 2
#define LED_PIN 13

unsigned long lastMotionTime = 0;
const unsigned long TIMEOUT = 10000; // 10 seconds

void setup() {
  pinMode(PIR_PIN, INPUT);
  pinMode(LED_PIN, OUTPUT);

  digitalWrite(LED_PIN, LOW);

  Serial.begin(9600);
```

```
  Serial.println("Real-Time Occupancy Detection System Started");
}

void loop() {

  bool motionDetected = digitalRead(PIR_PIN);

  if (motionDetected) {
    digitalWrite(LED_PIN, HIGH);
    lastMotionTime = millis();

    Serial.println("Occupancy Detected - Light ON");
  }

  if ((millis() - lastMotionTime) > TIMEOUT) {
    digitalWrite(LED_PIN, LOW);

    Serial.println("No Occupancy - Light OFF");
  }

  delay(100);
}
```

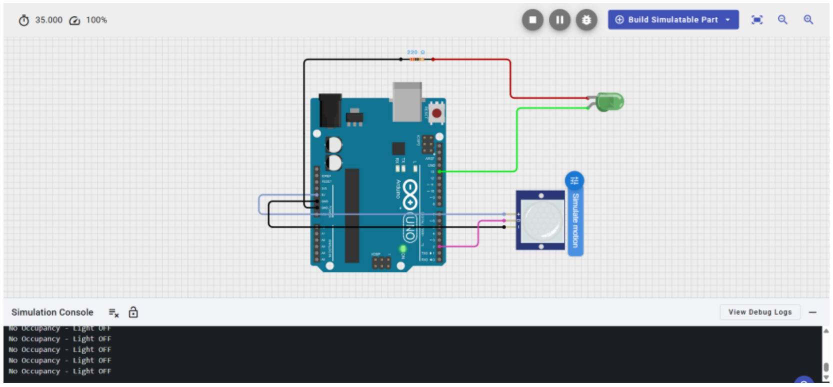
8. Results of Simulation

Test Case 1: No Occupancy

PIR Output = LOW

LED Status = OFF

Result = Successful

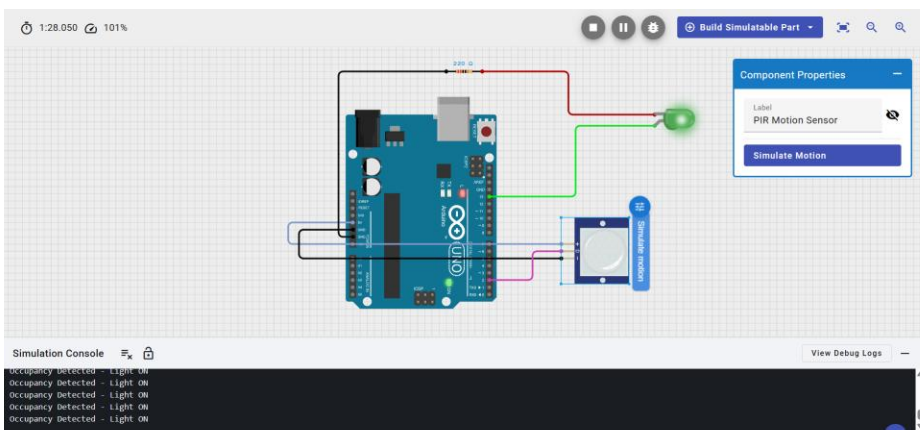


Test Case 2: Occupancy Detected

PIR Output = HIGH

LED Status = ON

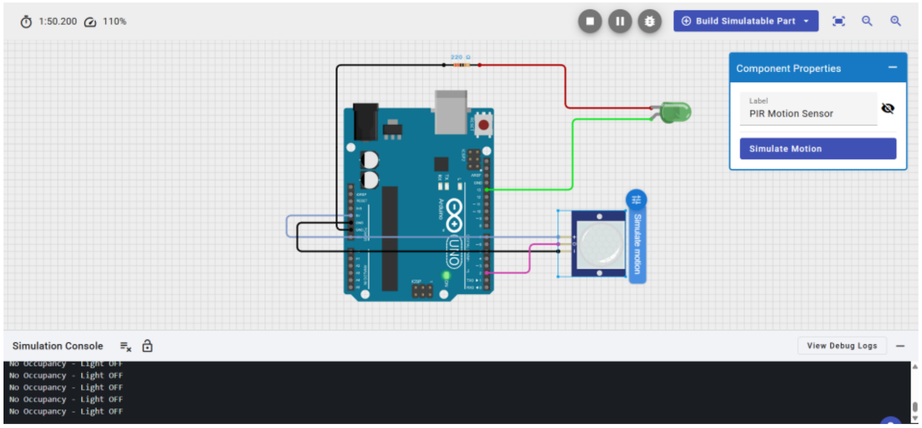
Result = Successful



Test Case 3: No Motion for 10 Seconds

LED Status = OFF

Result = Successful



Observation:

The system successfully detected human motion and controlled the lighting system automatically based on occupancy status, demonstrating effective energy-saving automation.

Test Case	PIR Output	LED Status	Result
No Occupancy	LOW	OFF	Successful
Occupancy Detected	HIGH	ON	Successful
No Motion for 10s	LOW	OFF	Successful

Estimated Energy Savings

Assuming:

- Room light power = 10W
- Light left ON unnecessarily for 5 hours daily

Energy wasted:

10 × 5 = 50 Wh/day = 0.05 kWh/day

Annual wastage:

0.05 × 365 = 18.25 kWh/year

The proposed system automatically turns OFF lighting when no occupancy is detected, thereby reducing unnecessary power consumption and improving energy efficiency.

9. Layout of System

System Layout

The proposed Real-Time Occupancy Detection System consists of an Arduino Uno microcontroller, a PIR motion sensor, and an LED-based lighting unit. The system is designed to monitor human occupancy in real time and automatically control lighting to improve energy efficiency.

Physical Layout

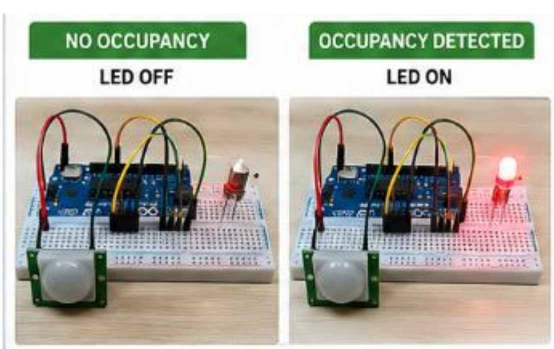
The PIR motion sensor is positioned at the entrance or within the monitored area to detect human movement. The sensor is connected to the Arduino Uno microcontroller, which acts as the central processing unit of the system. An LED is connected to the Arduino Uno to represent the room lighting system.

The PIR sensor continuously monitors the surroundings and sends motion detection signals to the Arduino Uno. When human motion is detected, the Arduino processes the sensor input and switches ON the LED. If no motion is detected for a predefined period, the Arduino automatically switches OFF the LED to conserve energy.

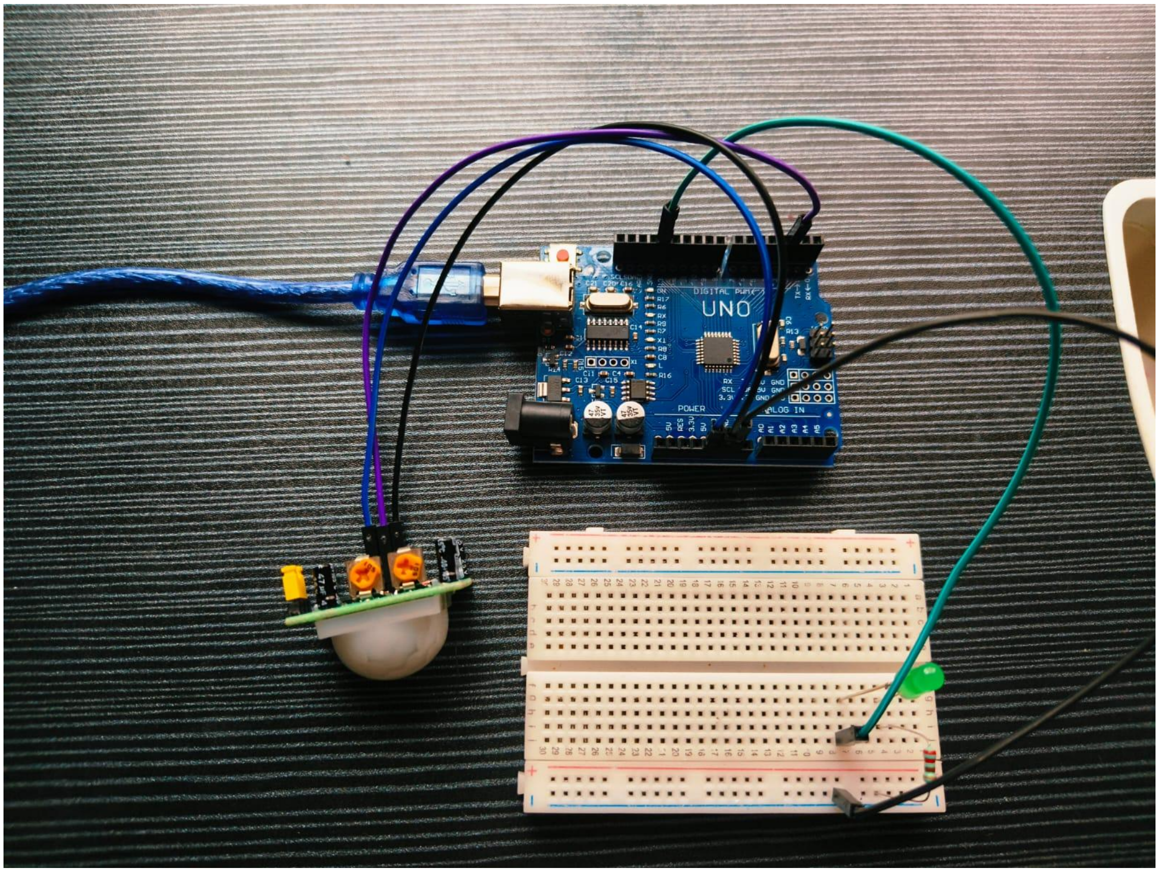
The components are assembled on a breadboard using jumper wires, enabling easy prototyping, testing, and modification of the circuit. The system operates as a standalone occupancy-based lighting control solution without requiring internet connectivity or external communication modules.

Working Flow

Human Motion
↓
PIR Sensor
↓
Arduino Uno
↓
LED ON/OFF Control

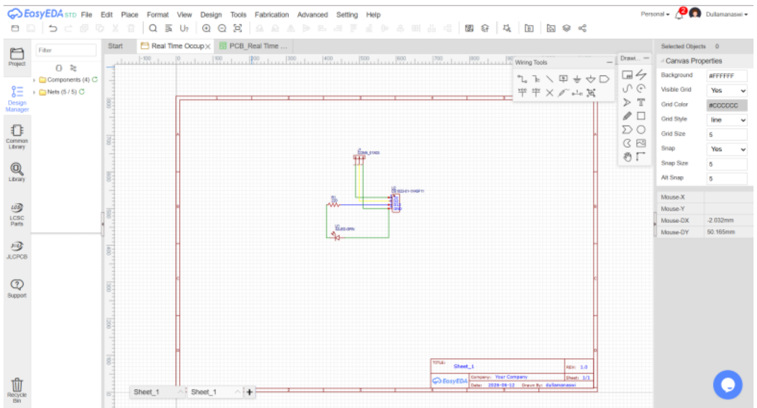


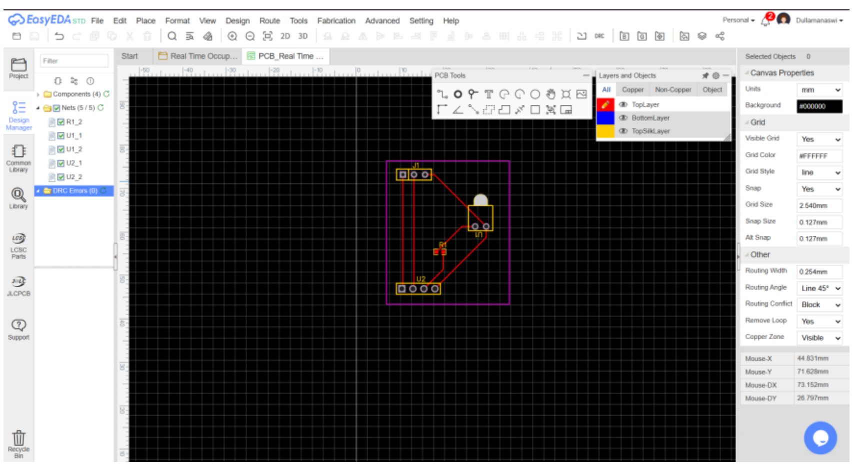
9.1 Hardware Prototype

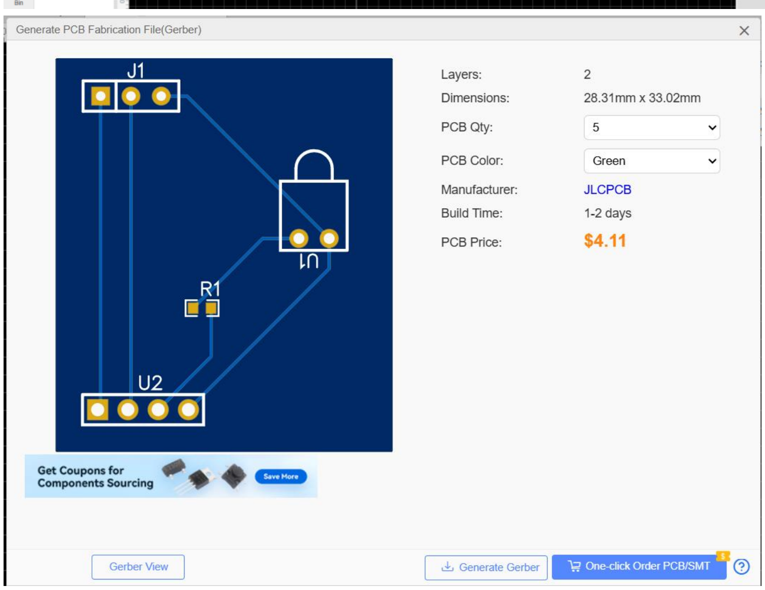


10. PCB Layout and Gerber Files

The generated Gerber files have been uploaded to the GitHub repository and can be directly used for PCB fabrication.







11. Video of the Demo

A demonstration video of the working prototype has been uploaded separately to the GitHub repository. The video shows motion detection, LED activation, timeout operation, and overall system functionality.

12. References

1. Arduino Uno Rev3 Technical Specifications, Arduino Documentation.

2. Arduino IDE Documentation and User Guide, Arduino Official Documentation.

3. U.S. Department of Energy, "Occupancy Sensors for Lighting Control Systems," Energy Saver Program Technical Guide.

4. HC-SR501 Passive Infrared (PIR) Motion Sensor Datasheet, Electronics Components Manufacturer Documentation.

5. ATmega328P Microcontroller Datasheet, Microchip Technology Inc.

6. National Institute of Standards and Technology (NIST), "Energy Efficiency and Smart Building Technologies," Technical Publications.

7. Simon Monk, Programming Arduino: Getting Started with Sketches, McGraw-Hill Education.

8. Michael Margolis, Arduino Cookbook, O'Reilly Media.